ANTARIKSA

Apple Developer Academy Candidate | Software Developer & Creative Technologist

Email: arik2002arik@gmail.com | GitHub: github.com/Antariksa21 | Location: Indonesia

01. EXECUTIVE STATEMENT & ACADEMY MOTIVATION

I am an aspiring iOS developer and data analyst driven by empathy, creative problem-solving, and human-centered design. With a strong foundation in transforming complex socio-health datasets into actionable visual insights (UNICEF Indonesia Stunting Dashboard), building native 2D game engines from scratch (Neon Runner), and engineering Apple-inspired glassmorphic user interfaces (NeoCalc), I bridge rigorous analytical thinking with refined aesthetic and interactive execution.

I am applying to the Apple Developer Academy to immerse myself in Challenge-Based Learning (CBL), master Swift & SwiftUI, and collaborate with multidisciplinary creators to architect transformative iOS applications that address pressing societal challenges in Indonesia.

02. CORE COMPETENCIES & TECHNICAL TOOLKIT

Languages & Fundamentals	Python, JavaScript (ES6+), HTML5, CSS3, SQL, OOP, Algorithmic Thinking
Data & Intelligence	Streamlit, Pandas, Plotly Express, Exploratory Data Analysis (EDA), Statistical Modeling, Caching
Frontend & Interactive	HTML5 Canvas 2D API, Web Audio API, Physics Simulation, Responsive Glassmorphism
Design & UI/UX	Apple Human Interface Guidelines (HIG), Micro-interactions, Accessibility, Clean Layouts
Methodologies & Teamwork	Challenge-Based Learning (CBL), Git/GitHub Version Control, Clean Architecture, Cross-Collaboration

03. FEATURED CASE STUDY 1 — PUBLIC HEALTH DATA INTELLIGENCE

UNICEF Indonesia: Interactive Nutrition & Stunting Analytics Dashboard

Role: Lead Data Analyst & Frontend Developer | Tech: Python, Streamlit, Pandas, Plotly Express

- The Challenge: Child stunting and malnutrition are critical public health crises across Indonesia. Public health datasets are often complex and tabular, making multi-variable trend discovery difficult for non-technical policymakers and field advocates.
- Engineering & Implementation: Architected a dynamic, web-based analytics dashboard featuring multi-dimensional filtering across indicators (Stunting, Wasting, Overweight, Exclusive Breastfeeding), gender, residence, and wealth quintiles with real-time reactive KPI updates.
- Socio-Demographic Insights: Uncovered strong negative correlation between maternal education level and stunting prevalence, and highlighted healthcare access disparities between urban and rural populations.
- Performance & Export: Integrated in-memory data caching (@st.cache_data) for instantaneous query execution and built an automated filtered dataset CSV exporter.

04. FEATURED CASE STUDY 2 — GAME ENGINE & CREATIVE ENGINEERING

Neon Runner: 2D Retro Cyber Synthwave Arcade Game Engine

Role: Creative Technologist & Game Developer | Tech: Vanilla JavaScript (ES6+), HTML5 Canvas 2D, Web Audio API

- The Challenge: Developing an engaging, 60 FPS endless runner game entirely from scratch without external game frameworks (Unity/Phaser), implementing native physics, particle simulation, and audio synthesis.
- Physics & Engine Engineering: Built a custom requestAnimationFrame game loop with state management, gravity, jump dynamics, multi-layered parallax backgrounds, and precise AABB (Axis-Aligned Bounding Box) collision detection.
- Audio & Particle Synthesis: Engineered real-time Canvas particle emitters (dust clouds, landing sparks, defeat explosion) and synthesized 8-bit sound effects using the native browser Web Audio API oscillators.
- Progression & Persistence: Designed procedural obstacle generation with adaptive speed curves and LocalStorage high-score persistence.

05. FEATURED CASE STUDY 3 — UI/UX & INTERACTIVE FRONT-END

NeoCalc: Premium Glassmorphic Calculator Application

Role: UI/UX Front-End Engineer | Tech: HTML5, CSS3 (Glassmorphism & Backdrop Filter), Vanilla JavaScript

- The Challenge: Creating an Apple-inspired desktop calculator that pairs sophisticated aesthetic elegance with robust mathematical accuracy and full keyboard accessibility.
- Design & Aesthetics: Engineered a responsive glassmorphic UI using CSS backdrop-filter blur, dynamic neon glow gradients, and tactile micro-interactions.
- Precision Engineering: Solved floating-point arithmetic precision issues (e.g., 0.1 + 0.2 = 0.3) with dedicated number sanitization and dynamic scientific notation for extreme values.
- Accessibility: Implemented bidirectional physical keyboard event listeners alongside touch/mouse interfaces for frictionless usability.

06. CHALLENGE-BASED LEARNING (CBL) & VISION FOR APPLE ACADEMY

VISION & COMMITMENT TO APPLE DEVELOPER ACADEMY
I aspire to bring my data-driven problem solving and creative engineering to develop high-impact native iOS and iPadOS applications. At the Apple Developer Academy, I aim to master Swift, SwiftUI, CoreML, and Apple HealthKit, collaborating across multidisciplinary teams within the Challenge-Based Learning (CBL) framework (Engage, Investigate, Act) to build world-class digital solutions for real community impact.